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**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

**COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards (BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile

65.5%

48

RUBRICS FOR CALCULATION

	Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
A	Problem Analysis	15% 3	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
B	Solution Design	20% 4	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
C	Technical Implementation	20% 4	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
D	Use of Tools & Technologies	10% 2	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
E	Problem-Solving Approach	10% 2	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
F	Innovation & Creativity	10% 2	Highly creative and original solution	Some creativity	Limited originality	No creativity
G	Testing & Validation	5% 1	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
H	Documentation & Presentation	5% 1	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
I	Teamwork / Collaboration	5% 1	Excellent coordination and	Good collaboration	Limited participation	No collaboration or contribution

65.5

	A	B	C	D	E	F	G	H	I	
Q ₁	2	2	2	1	1	2	1	0.5	1	12.5
Q ₂	2	2	3	1	1	2	1	0.5	1	13.5
Q ₃	2	3	2	2	2	2	0.5	1	0.5	15
Q ₄	2	2	2	2	2	1	0.5	1	0.5	13
Q ₅	2	2	2	1	1	1	1	1	0.5	11.5

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology.
(BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:50

Annexure A

Industry Problem-Solving Task

Code of Conduct:

I __j.manoj(192572143)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

j.manoj

Question 1:

A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

Introduction & Objective:

For a growing 20-person startup, network reliability, ease of management, and scalability are critical. The objective of this design is to deploy a cost-effective, high-performance local area network (LAN) that connects all workstations, printers, and mobile devices while ensuring seamless internet access and fault isolation.

Problem Analysis & Solution Design:

A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot.

- **Core Switching:** At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow.
- **Cabling Standards:** Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters—more than sufficient for any typical office floor plan.
- **Routing & Security:** The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation.
- **Wireless Extension:** A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices.

Network Diagram Representation (Star Topology):

Plaintext

[ISP / Internet]

|

[Router / Firewall] (pfSense / Cisco RV)

|

[Layer 2/3 Managed Switch] (Cisco Catalyst / Netgear 24-Port)

/ | | | \ \

W1 W2 W3 W4 ... Wi-Fi AP (Laptops/Mobiles)

Device & Cost Estimation Tables:

Device Name	Model / Specification	Quantity	Purpose
Managed Switch	Cisco Catalyst 24-Port Gigabit	1	Central hub connectivity
Router / Firewall	pfSense Appliance / Cisco RV Series	1	Internet gateway & NAT
Wireless AP	Wi-Fi 6 Access Point (PoE)	1	Wireless device extension
Cabling	Cat6 UTP Cable (per 305m box)	1	Physical transmission media

Advantages, Limitations & Scalability:

- Advantages:** Key advantage is fault isolation; if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.
- Limitations:** Dependency on the central switch; if the main switch fails, the entire network goes down.
- Future Scalability:** Easily scalable by adding a secondary switch via an uplink port or upgrading to a stacked switch configuration as the startup expands beyond 20 employees.

Conclusion:

The star topology provides the optimal balance of manageability, performance, and fault tolerance for a small startup environment.

Question 2:

A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

Introduction & Objective:

Banks demand near-zero downtime and maximum security, making network architecture resilience paramount. The objective is to design a fault-tolerant network infrastructure that eliminates single points of failure for core financial transactions.

Problem Analysis & Solution Design:

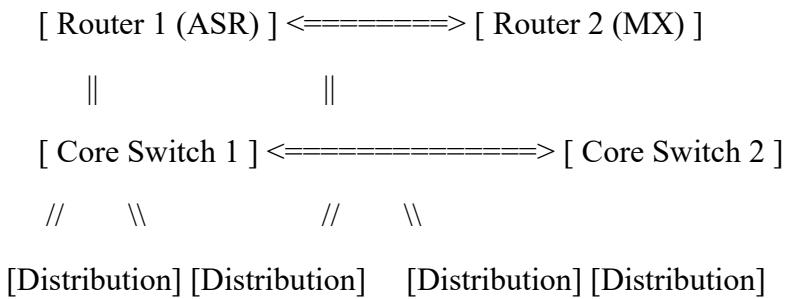
A bank demands near-zero downtime, making a full or partial mesh topology the right choice.

- **Topology Structure:** In a full mesh, every core network device—routers, core switches, and servers—connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches.
- **Routing & Redundancy:** Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design.

- **Transmission Media:** All links between core devices use fiber optic cables—either single-mode for long distances between buildings or multi-mode OM4 within the data center—because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions.

Network Diagram Representation (Partial/Full Mesh Core):

Plaintext



Specifications & Cabling Comparison Table:

Component	Enterprise Specification	Function in Banking Network
Core Routers	Cisco ASR / Juniper MX Series	BGP-based redundant internet routing
Fiber Cabling	Single-Mode & Multi-Mode OM4	Low-latency, EMI-immune high-speed backbone
Power Backup	Redundant PSUs & Enterprise UPS	Ensures zero downtime during electrical failures

Security, Uptime & Redundancy Analysis:

When a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

Conclusion:

The mesh architecture guarantees maximum availability and secure data transmission required in modern banking environments.

Question 3:

For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line

— the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

Introduction & Objective:

For educational environments operating under strict budget constraints, simplicity and low deployment costs are vital. The objective is to evaluate a traditional bus topology implementation for a small classroom lab.

Problem Analysis & Solution Design:

For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line—the bus—eliminating the need for a central switch or hub.

- **Cabling & Hardware:** A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection.
- **Contemporary Adaptation:** For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation—achieving bus-like simplicity at minimal cost.
- **Serverless Sharing:** No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

Network Diagram Representation (Bus Topology):

Plaintext

```
[Terminator]--[PC1]--[PC2]--[PC3]--[PC4]--[PC5]--[Terminator]
```

Hardware Requirements & Comparison Table:

Feature Component /	Bus Topology (Traditional/Modern)	Star Topology
Central Hardware	None (Shared coaxial backbone)	Central Switch / Hub required
Cost	Extremely Low	Moderate (Switch + extra cabling)
Fault Tolerance	Low (Single cable break disables all)	High (Isolated node failure)

Benefits, Drawbacks & Implementation Scenario:

- **Benefits:** Minimal hardware overhead and straightforward deployment.
- **Drawbacks:** The network is susceptible to collisions and a single cable break disrupts all connected machines.
- **Scenario:** Ideal for a supervised classroom environment with light usage (internet browsing, file sharing), where this trade-off is entirely acceptable.

Conclusion:

While outdated for large enterprises, a modified bus or low-cost shared-line setup serves as an economical teaching tool for basic lab environments.

Question 4:

A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is

the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

Introduction & Objective:

A large university campus spans multiple buildings with diverse networking demands. The objective is to design a scalable, hierarchical hybrid network architecture.

Problem Analysis & Solution Design:

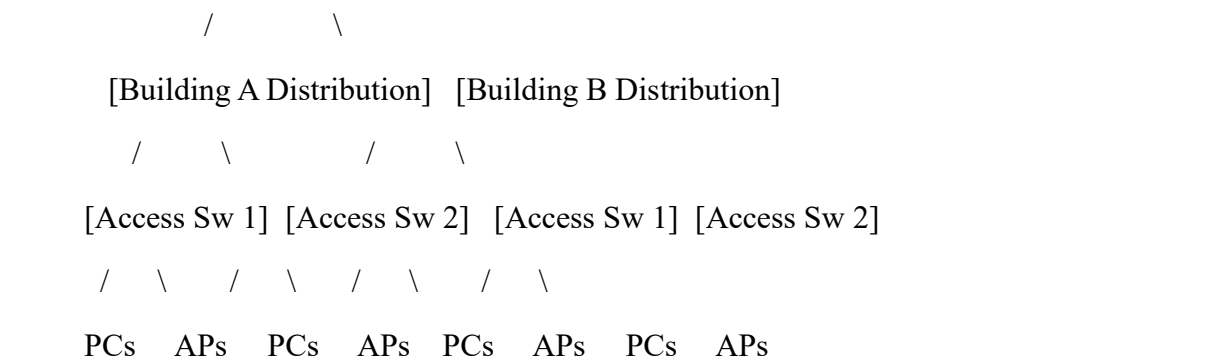
A large university campus serves thousands of users across multiple buildings—lecture halls, labs, libraries, dormitories, and administration—making a single topology inadequate.

- **Hybrid Combination:** A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency.
- **Campus Core & Backbone:** At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building.
- **Distribution & Access Layers:** Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables.
- **Hierarchical Model & Wireless:** This hierarchical three-layer model—core, distribution, and access—is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling.

Network Diagram Representation (Campus Hierarchical Architecture):

Plaintext

[Campus Core Router / Fiber Backbone (10/40 Gbps Ring)]



Device & Cabling Table:

Network Layer	Devices Deployed	Transmission Media
Core Layer	Multilayer Core Switches, Routers	Single-Mode Fiber (10/40 Gbps)
Distribution Layer	Building Distribution Switches	Dual Fiber Uplinks
Access Layer	Access Switches, PoE APs, Workstations	Cat6 UTP Cables

Conclusion:

The hybrid design successfully delivers redundancy at the backbone, simplicity at the building level, and seamless scalability campus-wide.

Question 5:

Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated

Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

Introduction & Objective:

Enterprise networks must segment traffic across various departments to optimize security and performance. The objective is to design a structured corporate network utilizing VLANs and inter-VLAN routing.

Problem Analysis & Solution Design:

Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security.

- **Departmental Access Layer:** Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department.
- **Core & Inter-VLAN Routing:** These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing.
- **Virtual LANs (VLANs):** Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic.
- **Security & Expansion:** A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

Network Diagram Representation (VLAN Architecture):

Plaintext

[Internet / Firewall Router]

|

[Layer 3 Core Switch (Inter-VLAN Routing)]

/ | \ \

[VLAN 10] [VLAN 20] [VLAN 30] [VLAN 40]

(HR Sw) (Finance Sw) (Sales Sw) (IT Sw)

Department-Wise Subnetting Table:

Department	VLAN ID	Subnet Range	Description
HR	VLAN 10	192.168.10.0/24	Human Resources Workstations
Finance	VLAN 20	192.168.20.0/24	Secure financial transactions
Sales	VLAN 30	192.168.30.0/24	Sales & CRM access
IT	VLAN 40	192.168.40.0/24	Administrative & Server management

Conclusion:

The VLAN-based multi-department architecture guarantees high security, organized broadcast domains, and effortless future expansion